Water Security Plan Report

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Date : 19/ 02/ 2024 Time : 6:23 PM User Name : balramkrishna

STATE: DELHI_7 » DISTRICT: NORTH_80 » BLOCK: ALIPUR_6076 » GRAM PANCHAYAT: HIRANKI_63897 » YEAR: 2023-24 UPDATE »

Gram Panchayat Information

State:	Delhi_7	District:	North_80
Block / Taluk :	Alipur_6076	GP Name :	Hiranki_63897
GP ID :	63897	Block Area (Ha) :	960.0000
GP area (Ha) :	321.5000	Watershed Code :	HIRA007
Watershed name :	Test2	Watershed area in Block / Taluk (Ha) :	100000
Sub-basin Code :	20	Sub basin name :	Yamuna Upper
Basin Code :	ss12	Basin Name :	Other

Location Map

Watershed Map

Aquifer Map

Ground Water Level

S.No.	Site Name	Location Details	Latitude	Longitude	Date	Well ID	Type of Well	Other Well Type	Ownership Type	Government Source	Diameter of well(m)	Depth of Well(mbmp)	Height of Measuring Point (mbgl)	Water Level (mbgl)	Water Level Date
1	Test gp	Hshdhdhs	28.64513	77.33247	2023-12-07	063897-021	Tube Well	null	Government	CGWB	36.0000	23.00	3.00	Dry	2023-12-07
	Test gp	Hshdhdhs	28.64513	77.33247	2023-12-07	063897-021	Tube Well	null	Government	CGWB	36.0000	23.00	3.00	23	2023-12-07
2	Newq	Ghazal	28.64515	77.3332	2023-07-19	063897-001	Dug Well		Government	CGWB	0.5000	15	1	Dry	2023-08-12
	Newq	Ghazal	28.64515	77.3332	2023-07-19	063897-001	Dug Well		Government	CGWB	0.5000	15	1	Dry	2023-07-25
	Newq	Ghazal	28.64515	77.3332	2023-07-19	063897-001	Dug Well		Government	CGWB	0.5000	15	1	Dry	2023-07-25
	Newq	Ghazal	28.64515	77.3332	2023-07-19	063897-001	Dug Well		Government	CGWB	0.5000	15	1	6	2023-07-25
	Newq	Ghazal	28.64515	77.3332	2023-07-19	063897-001	Dug Well		Government	CGWB	0.5000	15	1	12	2023-07-25
3	Yyy	Yyy	28.58429	77.23763	2023-08-01	063897-002	Piezometer		Private		0.2000	7	2.00	Dry	2023-10-13
	Yyy	Yyy	28.58429	77.23763	2023-08-01	063897-002	Piezometer		Private		0.2000	7	2.00	1	2023-09-04
	Yyy	Yyy	28.58429	77.23763	2023-08-01	063897-002	Piezometer		Private		0.2000	7	2.00	20	2023-08-12
4	Testes	Testes	28.64514	77.3325	2023-10-13	063897-003	Dug Well		Private		2.0000	5.00	2.00	Dry	2023-10-13
5	ivvivkv	jcivkvk	28.58428	77.2377	2023-10-16	063897-004	Piezometer		Private		0.5000	12.00	1.00	Dry	2023-10-22
6	Yeydye	Heyeheh	28.64508	77.33246	2023-11-01	063897-009	Dug Well		Private		2	3	6	Dry	2023-11-09
	Yeydye	Heyeheh	28.64508	77.33246	2023-11-01	063897-009	Dug Well		Private		2	3	6		2023-11-09

S.No.	Site Name	Location Details	Latitude	Longitude	Date	Well ID	Type of Well	Other Well Type	Ownership Type	Government Source	Diameter of well(m)	Depth of Well(mbmp)	Height of Measuring Point (mbgl)	Water Level (mbgl)	Water Level Date
	Yeydye	Heyeheh	28.64508	77.33246	2023-11-01	063897-009	Dug Well		Private		2	3	6		2023-11-02
	Yeydye	Heyeheh	28.64508	77.33246	2023-11-01	063897-009	Dug Well		Private		2	3	6		2023-11-02
	Yeydye	Heyeheh	28.64508	77.33246	2023-11-01	063897-009	Dug Well		Private		2	3	6		2023-11-02
	Yeydye	Heyeheh	28.64508	77.33246	2023-11-01	063897-009	Dug Well		Private		2	3	6	Dry	2023-11-02
	Yeydye	Heyeheh	28.64508	77.33246	2023-11-01	063897-009	Dug Well		Private		2	3	6	1.00	2023-11-02
	Yeydye	Heyeheh	28.64508	77.33246	2023-11-01	063897-009	Dug Well		Private		2	3	6	Dry	2023-11-01
7	Test2	Hxhhs	28.64513	77.33258	2023-11-24	063897-012	Dug-Cum-Bore well		Government	Others	0.2000	5.00	4.00	5	2023-11-24
	Test2	Hxhhs	28.64513	77.33258	2023-11-24	063897-012	Dug-Cum-Bore well		Government	Others	0.2000	5.00	4.00	4	2023-11-24
	Test2	Hxhhs	28.64513	77.33258	2023-11-24	063897-012	Dug-Cum-Bore well		Government	Others	0.2000	5.00	4.00	Dry	2023-11-24
8	Tetstts	Tetstts	28.64513	77.33239	2023-12-15	063897-026	Dug Well	null	Private	null	2	1.00	2.00	2	2023-12-15
9	Testsf	Testsf	28.64513	77.33244	2023-12-22	063897-028	Dug Well	null	Private	null	5.0000	4.00	3.00	3	2023-12-27
	Testsf	Testsf	28.64513	77.33244	2023-12-22	063897-028	Dug Well	null	Private	null	5.0000	4.00	3.00	Dry	2023-12-27
10	hello	hello	28.58431	77.23762	2024-01-03	063897-030	Piezometer	null	Private	null	0.2000	10.00	1.00	1	2024-01-03
11	Teststttsts	Teststttsts	28.64513	77.33247	2024-01-04	063897-031	Dug Well	null	Private	null	5.0000	4.00	3.00	Dry	2024-01-05
	Teststttsts	Teststttsts	28.64513	77.33247	2024-01-04	063897-031	Dug Well	null	Private	null	5.0000	4.00	3.00	2	2023-12-05

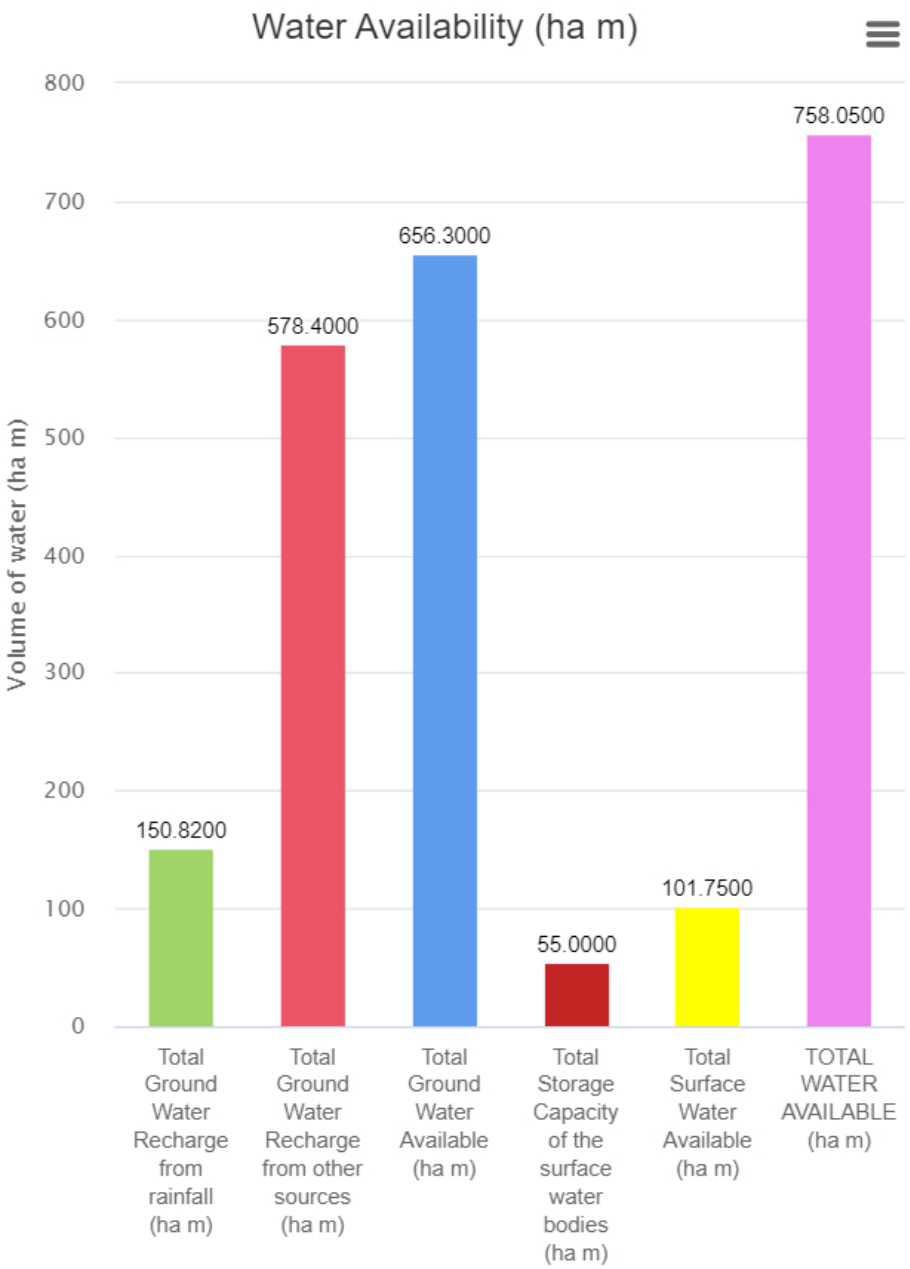
Ground Water Quality

S.No.	Village Name	Location	Latitude	Longitude	Year	Well Type	Source	Well ID
No Data								

S.No.	Well ID	Well Depth (mbgl)	Aquifer	PH	EC (µS/cm)	TDS (mg/L)	TH (mg/L)	Ca (mg/L)	Mg (mg/L)	Na (mg/L)	K (mg/L)	Carbo nate (mg/L)	Bicar bonate (mg/L)	Sulph ate (mg/L)	Chlor ide (mg/L)	Fluo ride (mg/L)	Nitr ate (mg/L)
No Data																	

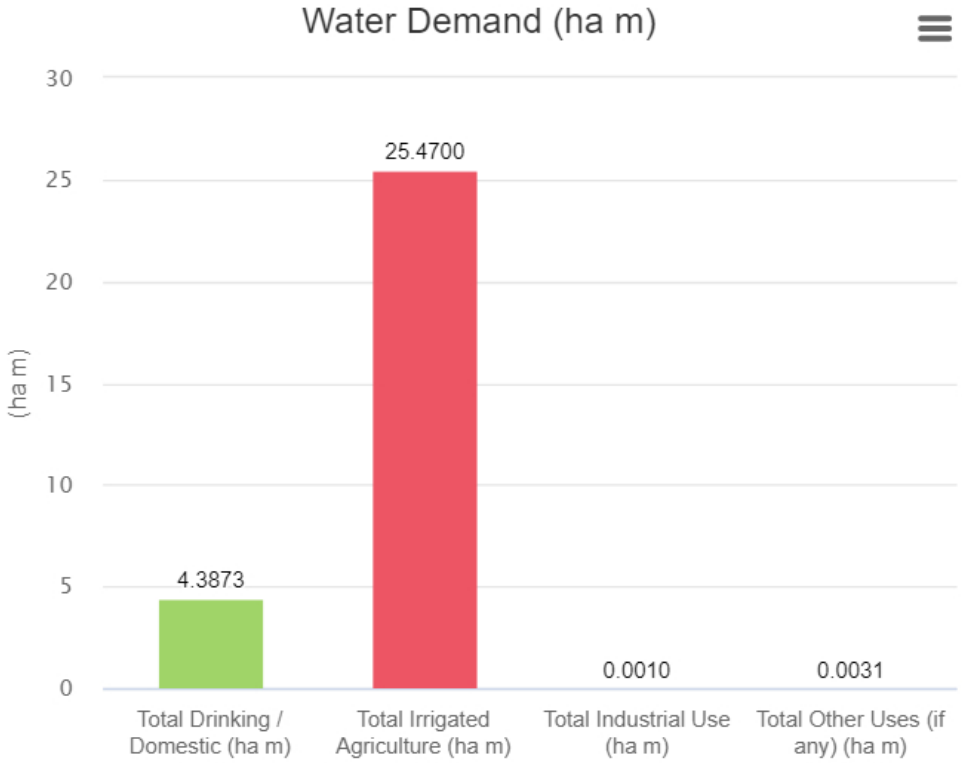
Water Availability Report

Water Availability		
1.	Total Available Water From Rainfall	
a.	Annual Rainfall (mm)	405.0000
b.	Estimated Volume of Rainfall (ha m)	130.2100
c.	Evapotranspiration rate (mm)	
d.	Estimated Evapotranspiration losses (ha m)	
e.	Estimated Water Available From Rainfall (ha m)	
f.	Weighted Average Runoff Coefficient	0.2000
g.	Rainfall Runoff(ha m)	
2.	Ground Water	
a.	Recharge From Rainfall (ha m)	
(i)	Monsoon	139.5400
(ii)	Non-Monsoon	11.2800
	Total Ground Water Recharge from rainfall (ha m)	150.8200
b.	Ground Water Recharge from other sources (ha m)	
(i)	Monsoon	263.2100
(ii)	Non-Monsoon	315.1900
	Total Ground Water Recharge from other sources (ha m)	578.4000
c.	Losses Due to Natural Discharge (ha m)	72.9200
	Total Ground Water Available (ha m)	656.3000
d.	Ground Water Withdrawal (ha m)	642.5300
3.	Surface Water	
a.	Surface Water available in storages	
(i)	Number of Surface Water Bodies and Storage Capacity (ha m)	
Type of Structure	No. of Structures	Storage Capacity (ha m)
Ponds	55	55
	Total Storage Capacity of the surface water bodies (ha m)	55.0000
(ii)	Losses if Any (%)	15.0000
(iii)	Surface Water Available in Storages (ha m)	46.7500
b.	Transported Water from outside GP area (ha m)	55.0000
c.	Total Surface Water Available (ha m)	101.7500
4.	TOTAL WATER AVAILABLE	
	TOTAL WATER AVAILABLE (ha m)	758.0500



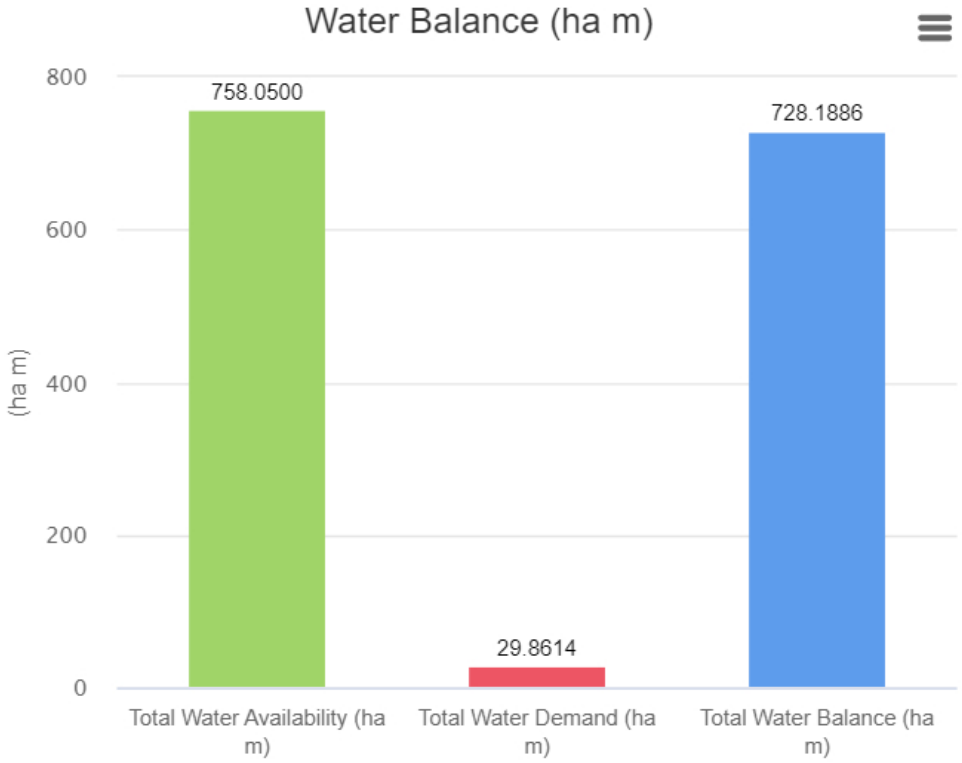
Water Demand Report

Water Demand		
A	Drinking /Domestic (ha m)	
i)	From Surface Water	3.7292
ii)	From Ground Water	0.6581
	Total Drinking /Domestic (ha m)	4.3873
B	Irrigated Agriculture (ha m)	
i)	From Surface Water	14.0085
ii)	From Ground Water	11.4615
	Total Irrigated Agriculture (ha m)	25.4700
C	Industrial Use (ha m)	
i)	From Surface Water	0.0003
ii)	From Ground Water	0.0007
	Total Industrial Use (ha m)	0.0010
D	Other Uses (if any) (ha m)	
i)	From Surface Water	0.0029
ii)	From Ground Water	0.0002
	Total Other Uses (if any) (ha m)	0.0031



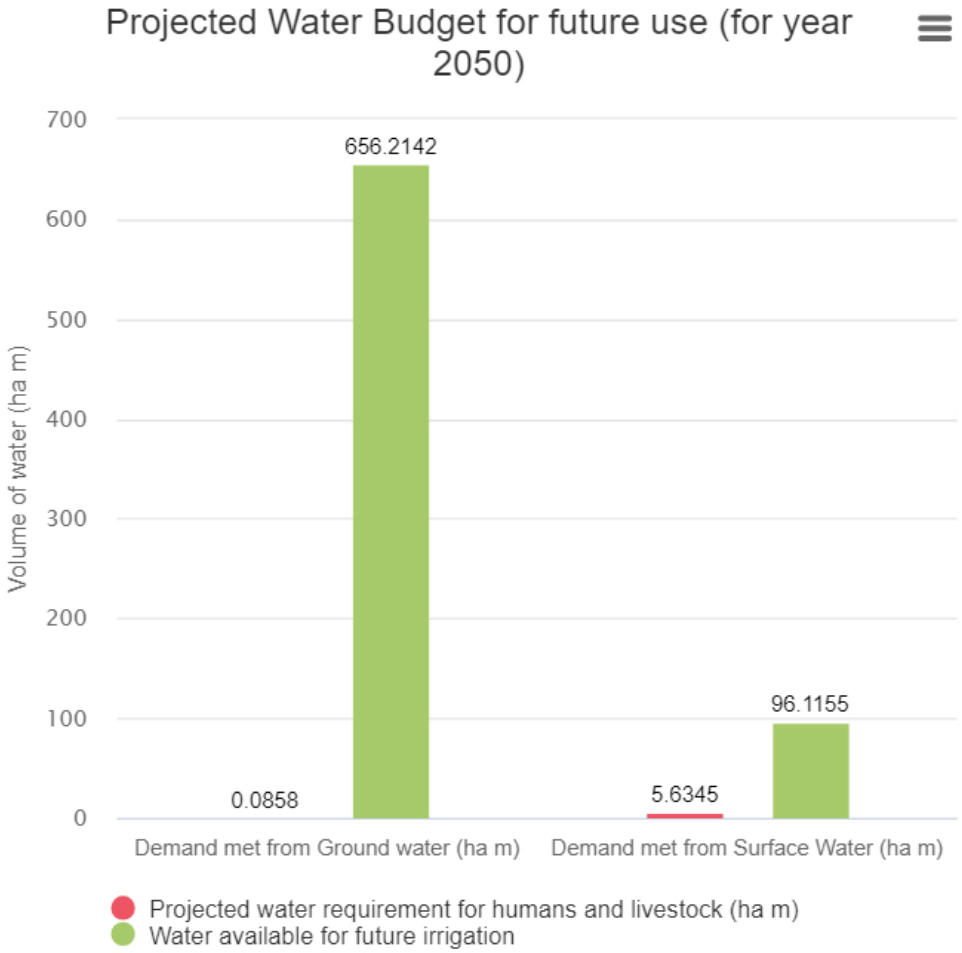
Water Balance Report

Water Balance		
A	Water Availability (ha m)	
i)	Total Ground Water Available	656.3000
ii)	Total Surface Water Available	101.7500
	Total Water Availability (ha m)	758.0500
B	Water Demand (ha m)	
i)	Total Surface Water Demand	17.7409
ii)	Total Ground Water Demand	12.1205
	Total Water Demand	29.8614
C	Balance [Surplus (+) / Deficit (-)]	
i)	Total Ground Water Balance	644.1795
ii)	Total Surface Water Balance	84.0091
iii)	Total Water Balance	728.1886



Water Budget Report

A	Allocation for Drinking & Domestic Requirements	
I	Humans	
i)	Present Human Population	1950.0000
ii)	Annual population growth rate (r)	1.0000
iii)	No. of years	28.0000
iv)	Projected human population (as in 2050)	2551.0000
v)	Per capita requirement (L)	60.0000
vi)	Projected water requirement for humans (as in 2050) (ha m)	5.5867
II	Livestock	
i)	Present livestock population	500.0000
ii)	Annual population growth rate (r)	0.5000
iii)	No. of years	28.0000
iv)	Projected livestock population (as in 2050)	572.0000
v)	Average per capita requirement (L)	6.4000
vi)	Projected water requirement for livestock (as in 2050) (ha m)	0.1336
B	Projected water requirement for humans and livestock (ha m) (as in 2050)	
	Total Projected water requirement for humans and livestock (as in 2050) (ha m)	5.7203
i)	Proposed share to be met from Ground water (in %)	1.5000
ii)	Proposed share to be met from Surface water (in %)	98.5000
iii)	Demand met from Ground water (ha m)	0.0858
iv)	Demand met from Surface water (ha m)	5.6345
C	Water available for future irrigation	
i)	Ground Water	656.2142
ii)	Surface water	96.1155
iii)	Total water available for future irrigation	752.3297

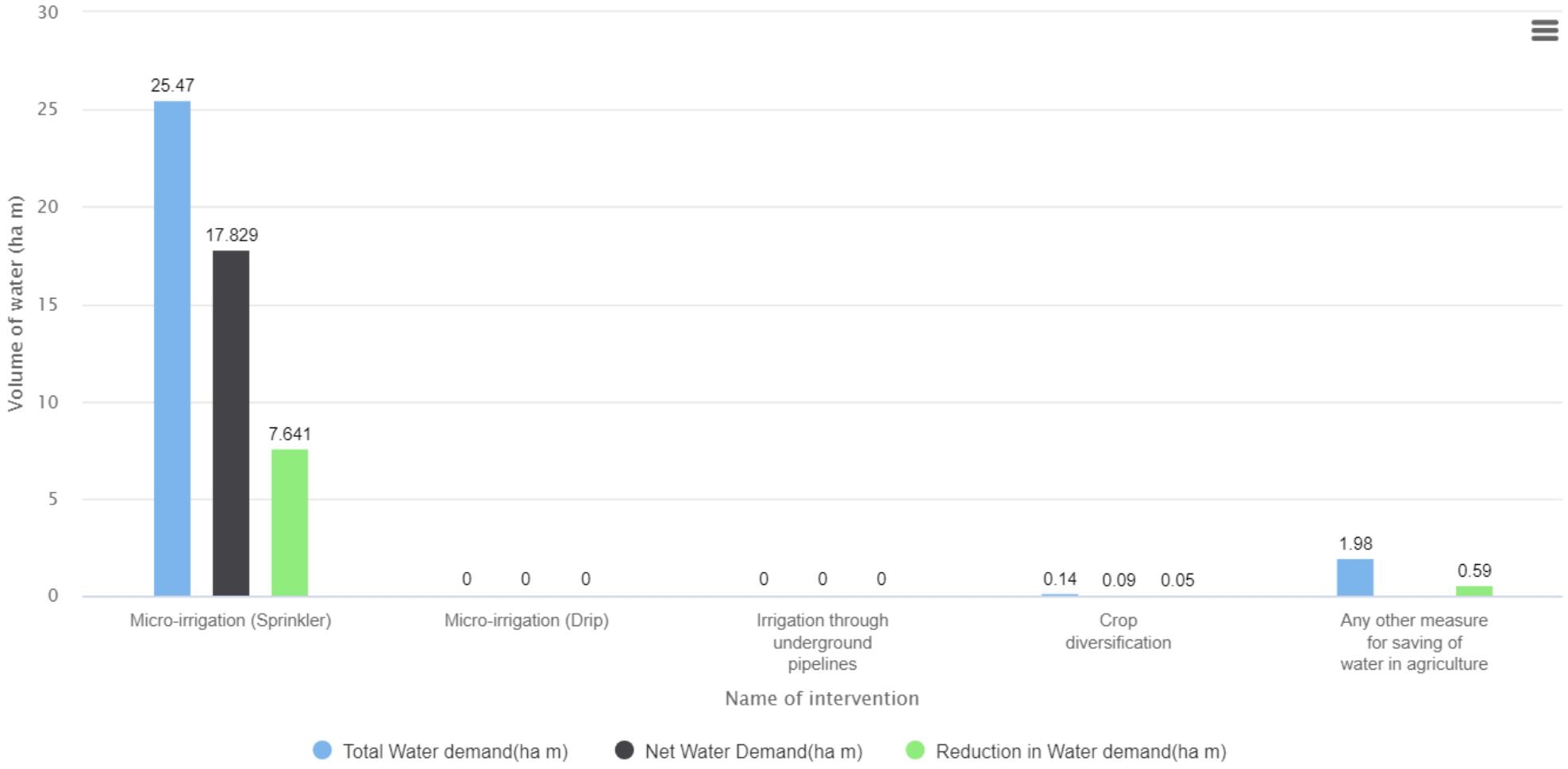


Demand Side

Total Area under Conventional Irrigation (ha) : 45

Demand Side						
Sr No	Name of Intervention	Proposed Area(ha)	Total Water demand(ha m)	Net Water Demand(ha m)	Reduction in Water demand(ha m)	Approximate cost(in lakh)
1	Sprinkler	45.0000	25.4700	17.8290	7.6410	44.0000
2	Drip	0.0000	0.0000	0.0000	0.0000	0.0000
3	Pipelines	0.0000	0.0000	0.0000	0.0000	0.0000
4	Crop Diversification	10	0.1400	0.0900	0.0500	45.0000
5	Other Measures	45.0000	1.9800	--	0.5900	33.0000
	Total	100.0000	27.5900	17.9190	8.2810	122.0000

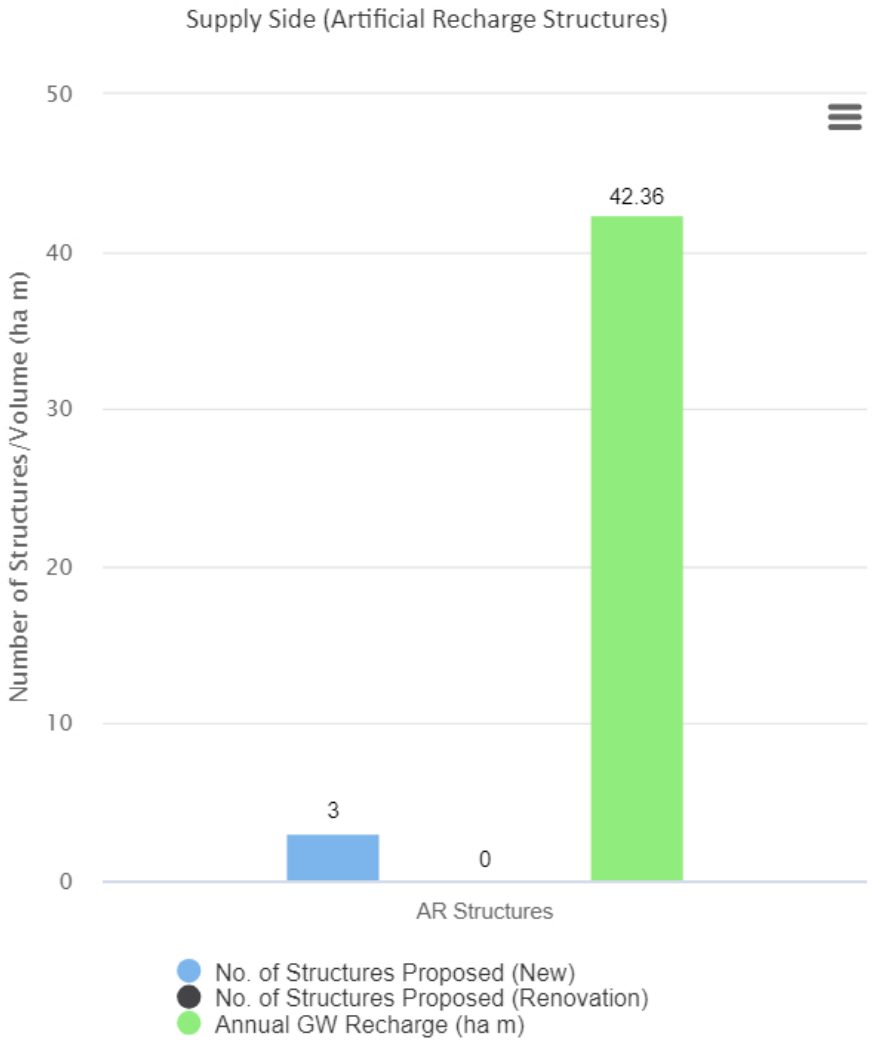
Funding Source						
Sr No	Name of Intervention	Area Proposed (ha)	Financial Year	Department Name	Scheme Name	Approximate Cost(in Lakh)
1	Sprinkler	45	2024-25	Department of Agriculture	GPDC 15th Finance Commission	44
2	Crop Diversification	10	2024-25	Department of Agriculture	Mera Pani Meri Virasat(MPMV)	45
3	Other Measures	45	2024-25	Department of Agriculture	Jal Jeevan Mission	33
	Total	100				122.0000

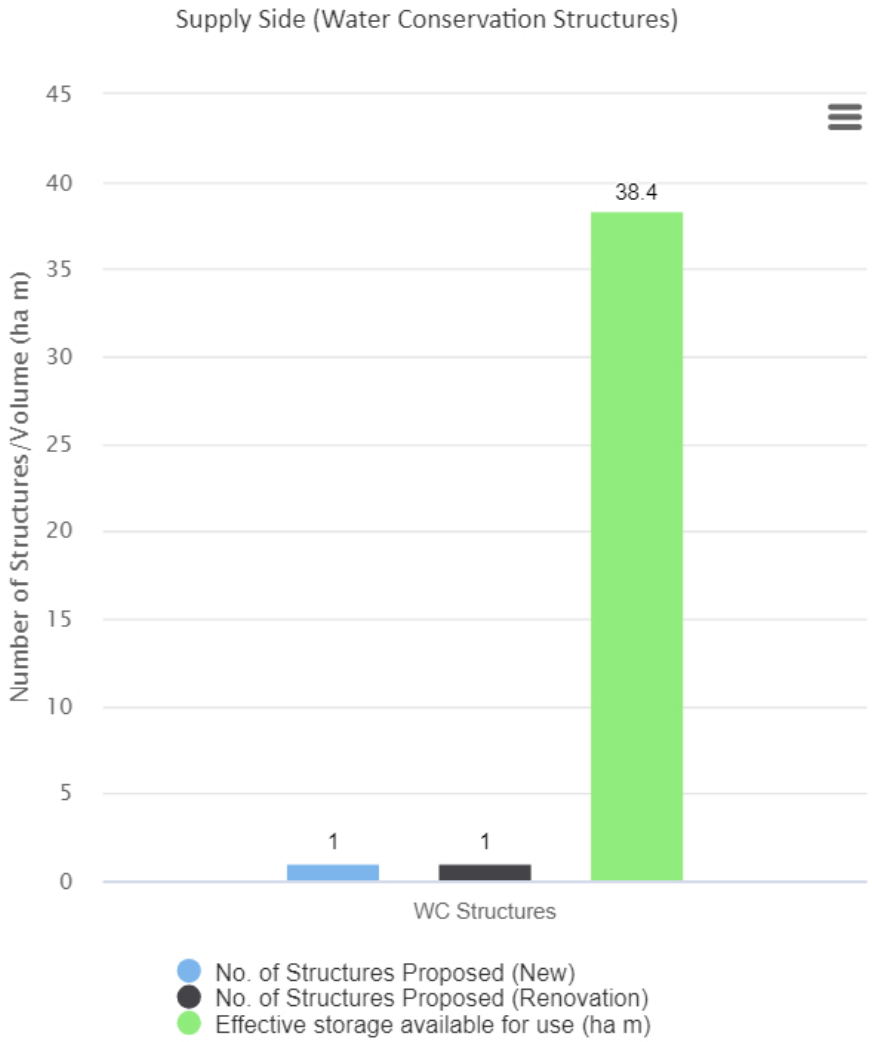

30
25
20
15
10
5
0
Volume of water (ha m)
25.47
17.829
7.641
0
0
0
0
0
0
0.14
0.09
0.05
1.98
0.59
Micro-irrigation (Sprinkler)
Micro-irrigation (Drip)
Irrigation through underground pipelines
Crop diversification
Any other measure for saving of water in agriculture
Name of intervention
Total Water demand(ha m)
Net Water Demand(ha m)
Reduction in Water demand(ha m)

Supply Side

Supply Side (Artificial Recharge Structures)										
SR No	Name of Structure	Storage Capacity(ha m)	Annual GW Recharge(ha m)	Village Name	Latitude	Longitude	Financial Year	Department Name	Scheme Name	Approximate Cost (in Lakhs)
1	Basin / Percolation Tanks(New)	100	40.00	Hiranki	28.582864	77.234457	2023-24	Minor Irrigation Department	Gray Water Management-Phase-II	20
2	Dug Well Recharge(New)	10	2.0000	Hiranki	72.36	23.56	2023-24	Department of Forest	Jal Jeevan Mission	2
3	Stop Dam(New)	3	0.3600	Hiranki	34.34	56.45342	2023-24	Water Resources Department	Other - Test	23
	Total	113.0000	42.3600							45.0000

Supply Side (Water Conservation Structures)										
SR No	Name of Structure	Storage Capacity(ha m)	Effective storage available for use (ha m)	Village Name	Latitude	Longitude	Financial Year	Department Name	Scheme Name	Approximate Cost (in Lakhs)
1	Farm Pond(Renovation)	300	18.00	Hiranki	28.581964	77.233921	2023-24	Panchayat Raj Engineering Department	Jal Jeevan Mission	34
2	Farm Pond(New)	34	20.4000	Hiranki	23.454	23.3534	2023-24	Irrigation & Water Resource Development	Other - Test WC	45
	Total	334.0000	38.4000							79.0000


Supply Side (Artificial Recharge Structures)
50
40
30
20
10
0
Number of Structures/Volume (ha m)
42.36
3
0
AR Structures
No. of Structures Proposed (New)
No. of Structures Proposed (Renovation)
Annual GW Recharge (ha m)


Supply Side (Water Conservation Structures)
45
40
35
30
25
20
15
10
5
0
Number of Structures/Volume (ha m)
38.4
1
1
WC Structures
No. of Structures Proposed (New)
No. of Structures Proposed (Renovation)
Effective storage available for use (ha m)

PGWM Committee

PGWM Committee (VWSC/WUA/CBO)				
Sr.no	Name	Designation	Gender	Mobile /Landline No.
1	wrewre	ewrer	Male	6453434545

Details of PGWM Meeting in which resolution was passed												
Sr.no	State	District	Block	Gram Panchayat	Date	GP Name	Village Name	Meeting Location	Male	Female	Transgender	Total number of attendees
1	Delhi_7	North_80	Alipur_6076	Hiranki_63897	2024-01-30	Hiranki_63897	Hiranki	223	34	23	4	61
2	Delhi_7	North_80	Alipur_6076	Hiranki_63897	2024-02-19	Hiranki_63897	Hiranki	223	3	34	34	71

Photograph

Govt. of India
Ministry of Jal Shakti
Dept. of Water Resources, River Development
& Ganga Rejuvenation





Govt. of India
Ministry of Jal Shakti
Dept. of Water Resources, River Development
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आज़ादी का अमृत महोत्सव

ATAL BHUJAL YOJANA

G20

